

TECHNICAL DESIGN SPECIFICATION

Hardware Implementation of Analog Matrix Cell (2T1C) for Inverted MAML

Document Type: Specification & Calculation Basis | **Target Components:** DOX3134A, MCP6024T | **Status:** Reviewed and Optimized (English Translation)

1. Introduction and System Overview

This document summarizes the critical design parameters and calculation basis for the core block of the analog compute matrix, based on a **2T1C** cell (two transistors, one capacitor). The architecture interacts with a digital control unit executing an **Inverted Meta-Learning (MAML)** algorithm. The purpose of this algorithm is to dynamically compensate for inherent hardware non-idealities, such as voltage leakage (droop) and component variations, in real-time during ongoing inference.

2. Timing and Sampling Dynamics

The system's time constants are asymmetrically divided to separate the high-frequency compute path (payload inference) from the low-frequency calibration and refresh cycle.

- **Refresh Cycle (10 ms):** The time interval within which the analog weights on the matrix cell capacitors must be updated to prevent uncontrolled discharge.
- **Refresh Window ($\leq 100 \mu\text{s}$):** The dedicated charging and adjustment time allocated per row/cell. This window constitutes at most 1/100th of the total refresh cycle to minimize system downtime.
- **Sampling Time (2 μs):** The throughput time for the analog matrix multiplication per compute sample during ongoing inference.

3. Voltage Levels and Operating Regions (Triode Condition)

For the cell's MOSFET to function as an analog multiplier, it must be forced to operate in the **linear region (triode region)**, where the channel acts as a voltage-controlled resistor. If the transistor is allowed to enter the saturation region, the current levels off and the multiplication capability is lost. The fundamental condition is:

$$V_{ds} < V_{gs} - V_{th}$$

Given a fixed Source potential of $V_s = 1.65 \text{ V}$ (the system's virtual ground reference) and a typical threshold voltage of $V_{th} \approx 0.6 \text{ V}$ for the **DOX3134A** transistor, the voltage swings have been optimized to handle a voltage drop (droop) of approximately 10% during the cycle.

Data Swing on the Row Line (250 mV Swing)

By limiting the row swing from **1.65 V** (Digital 0) to **1.90 V** (Digital 255), a maximum $V_{ds} = 0.25 \text{ V}$ is established. This narrow swing guarantees that the transistor remains securely within the linear region, even when the weight voltage on the Gate has dropped to its lowest point at the end of a refresh cycle.

Weight Swing on the Gate (1.0 V Swing)

The weight voltage is allowed to swing between **2.1 V** (Weight 0, set precisely near the threshold voltage's effective limit during droop) up to **3.1 V** (Weight 255). At a medium weight (originally **2.6 V**), the voltage drops by approximately **0.1 V** to **2.5 V** just before the next refresh. This yields $V_{gs} = 2.5 \text{ V} - 1.65 \text{ V} = 0.85 \text{ V}$, which establishes a linear ceiling of $V_{gs} - V_{th} = 0.85 \text{ V} - 0.6 \text{ V} = 0.25 \text{ V}$. Since $V_{ds} \leq 0.25 \text{ V}$, the triode condition is maintained precisely at the boundary.

4. RC Network and Component Dimensioning

To avoid extreme resistance values on the printed circuit board (PCB) and maintain high immunity to environmental noise and parasitic leakage currents, a robust and stable RC landscape has been calculated.

Discharge Network (Droop Control)

By selecting a local weight capacitor of $C = 100 \text{ nF}$ and a discharge resistor of $R = 1 \text{ M}\Omega$ tied to ground, the discharge time constant is calculated as:

$$\tau_{\text{discharge}} = 1 \text{ M}\Omega \cdot 100 \text{ nF} = 100 \text{ ms}$$

The voltage drop over the refresh cycle of $t = 10 \text{ ms}$ is governed by:

$$V(t) / V_0 = e^{-10\text{ms} / 100\text{ms}} = e^{-0.1} \approx 0.9048$$

This results in a deterministic and controlled voltage drop of exactly **~9.5%** (approx. 0.1 V), which matches the system architecture perfectly since the algorithm's snapshot windows compensate for this known decay curve.

Charging Network and Stability

To drive the relatively large capacitive load (**100 nF**) using the **MCP6024T** operational amplifier, a series isolation resistor of $R_s = 100 \text{ }\Omega$ is introduced at the OP-amp's output. This isolates the capacitive load, completely eliminating the risk of resonance or self-oscillation in the feedback loop, thereby ensuring phase stability.

The time constant for charging is $\tau_{\text{charge}} = 100 \text{ }\Omega \cdot 100 \text{ nF} = 10 \text{ }\mu\text{s}$. To reach an acceptable settling level of approximately 95% of the intended voltage step, it requires:

$$3 \cdot \tau_{\text{charge}} = 30 \text{ }\mu\text{s}$$

Since $30\ \mu\text{s}$ is well within the maximum allowed refresh window of $100\ \mu\text{s}$, the driving stage successfully charges the cell to the intended target value without lagging.

System Architectural Notice Regarding Droop Tolerance

In a conventional analog matrix, a voltage drop of $\sim 9.5\%$ would cause severe calculation drift. Within this hybrid architecture, however, this is perfectly acceptable. The Inverted MAML algorithm utilizes short, temporally fixed measurement windows (snapshots) placed in identical phase positions across the refresh cycles, making the time-dependent voltage drop completely deterministic and fully compensable.

5. Summary Table: System and Hardware Parameters

Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	$\leq 100\ \mu\text{s}$	Time allocated to charge the cell (95% reached in $\sim 30\ \mu\text{s}$).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	$\sim 9.5\%$	Controlled voltage drop ($\sim 0.1\ \text{V}$) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.